

Romil Prince Bansal

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EDUCATION

Thapar Institute of Engineering and Technology
Bachelor of Engineering in Computer Science and Engineering

Patiala, Punjab, India
Expected June 2026

PROJECTS

Drowsiness Detection System | *Python, OpenCV, TensorFlow, CNN, Computer Vision*

- Built a **real-time driver alertness monitoring system** leveraging OpenCV Haar Cascade classifiers for facial and ocular region localization, paired with a custom **CNN trained on 4,000 MRL Eye Dataset samples** (2,000 open + 2,000 closed), attaining **95%+ validation accuracy**.
- Architected a **3-block convolutional network** incorporating BatchNormalization, Dropout regularization, and data augmentation across **6 simulated lighting environments** — optimized via Adam with EarlyStopping and ReduceLROnPlateau callbacks.
- Applied **TFLite model quantization** and parallelized preprocessing pipelines, elevating inference throughput by **30%** (23 FPS → 30 FPS) with no accuracy degradation, achieving sub-100ms alert latency.
- Deployed **continuous video stream processing** with a rolling-window fatigue scorer and pygame-based audio alerts — activating warnings upon 15 consecutive closed-eye frames, cutting simulated fatigue incidents by **40%** across 500+ test scenarios.

AI Caption Agent | *Python, FastAPI, Groq API, Llama 4 Vision, React*

- Developed and deployed a **full-stack agentic AI platform** using FastAPI and Groq's Llama 4 Vision model, automating multi-platform social media caption generation from image uploads with **sub-10s end-to-end latency**.
- Constructed a **multimodal vision pipeline** extracting 10-dimensional content briefs (subject, mood, aesthetic, category) and routing outputs through **6 platform-specific prompt templates** (Instagram, YouTube, Twitter/X, LinkedIn, Hashtags, Alt Text).
- Delivered a **RESTful API** with modular route architecture (upload, analyze, generate-captions) and a responsive dark-themed React dashboard featuring drag-and-drop upload, tone selection, and one-click copy.
- Enforced **production-grade reliability** via chunked JSON parsing, structured error handling, and environment-based API key management — supporting 5+ caption formats with configurable tone profiles.

Retail & E-Commerce Analytics Platform | *Python, Streamlit, XGBoost, MLxtend, TextBlob, Plotly*

- Developed a **unified ML analytics dashboard** consolidating three modules — Market Basket Analysis, Customer Lifetime Value (CLV) prediction, and NLP-driven Sentiment Demand Analysis — into a single Streamlit interface.
- Deployed the **Apriori algorithm** for association rule mining across 500,000+ retail transactions, surfacing product co-occurrence insights with tunable support, confidence, and lift thresholds for cross-selling optimization.
- Trained an **XGBoost regression model** on RFM (Recency, Frequency, Monetary) and demographic features to forecast CLV, yielding high R^2 scores with automated 4-tier segmentation (Low, Mid, High, Premium).
- Engineered a **TextBlob NLP pipeline** for bulk sentiment classification of customer reviews — generating polarity scores, demand signals, and interactive word clouds while processing **10,000+ reviews** via chunked streaming.

TECHNICAL SKILLS

Languages: Python, C, C++, HTML5, CSS3, SQL

Frameworks & Libraries: TensorFlow, Scikit-Learn, XGBoost, MLxtend, NumPy, Pandas, OpenCV, Matplotlib, Plotly, TextBlob, Streamlit, FastAPI, React, Bootstrap, Tailwind CSS

Tools & Platforms: Git, GitHub, Tableau, Jupyter Notebook, Excel

Databases: MySQL

APIs & Services: Groq API, Llama 4 Vision, pygame

Concepts: Machine Learning, Deep Learning, Computer Vision, NLP, Data Structures & Algorithms, Statistical Analysis, ETL Pipelines, Data Visualization, DBMS, Operating Systems, Agile/Scrum

LEADERSHIP AND VOLUNTEERING

Mudra — Cultural Club

Core Member

January 2023 – May 2024

Thapar Institute of Engineering and Technology

- Spearheaded logistics and on-ground execution for **5+ large-scale cultural events** (500+ attendees each), coordinating cross-functional teams of **20+ volunteers**.
- Oversaw backstage operations and audience engagement strategies, contributing to a **30% improvement** in participant satisfaction scores across events.